



Mortar - Types, Ingredients, Proportions & Suitability.

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Introduction to Mortar

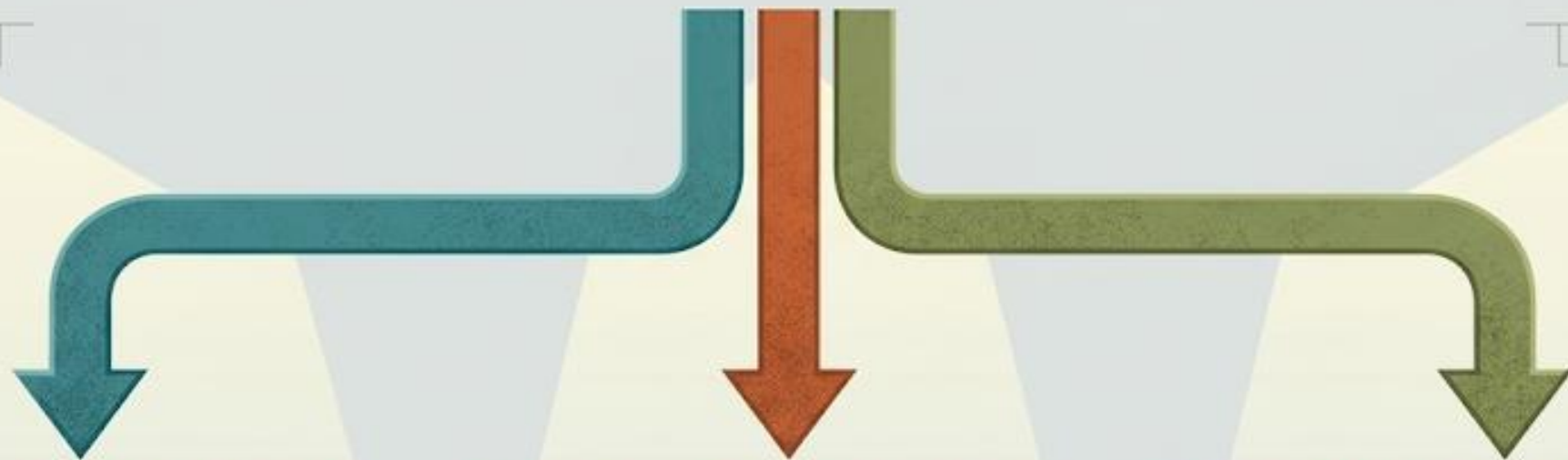
- ❖ Mortar is a workable paste used in construction to bind building units such as bricks, stones, and concrete blocks together.
- ❖ It fills the gaps between these units, provides structural stability, and creates a strong, durable bond.
- ❖ Mortar also helps distribute loads evenly, improves the appearance of masonry work, and prevents the entry of water and air through joints.
- ❖ Mortar is made by mixing a binding material (such as cement or lime), fine aggregate (sand), and water.
- ❖ Depending on the construction requirements, additives or admixtures may also be included to improve properties such as workability, strength, or water resistance.



Functions of Mortar

- ❖ Mortar binds bricks, stones, and concrete blocks together to form a strong structure.
- ❖ Mortar improves the appearance of masonry by providing neat and uniform joints.
- ❖ Mortar protects masonry from weathering and environmental damage.
- ❖ Mortar accommodates small movements caused by temperature changes and settlement.
- ❖ Mortar provides a smooth base for plastering, pointing, and other finishing works.
- ❖ Mortar improves the fire resistance of masonry structures.
- ❖ Mortar helps reduce the transfer of heat and sound through walls.
- ❖ Mortar is used for repairing cracks and damaged masonry.
- ❖ Mortar increases the overall strength, durability, and service life of masonry structures.

CLASSIFICATION OF MORTAR



BASED ON BINDING MATERIAL

- ✓ Lime mortar
- ✓ Surkhi mortar
- ✓ Cement mortar
- ✓ Gauged mortar
- ✓ Gypsum mortar

BASED ON BULK DENSITY

- ✓ Heavy weight mortar
- ✓ Light weight mortars

BASED ON NATURE OF APPLICATION

- ✓ Masonry mortar
- ✓ Finishing mortar





Classification of Mortar

- ❖ Heavy mortar is a type of mortar with a bulk density of 15 kN/m^3 (1500 kg/m^3) or more, prepared using dense fine aggregates such as heavy quartz sand, crushed stone sand, or other heavy mineral aggregates.
- ❖ It has high compressive strength, good durability, high resistance to wear, and better load-bearing capacity, making it suitable for heavy structural and masonry work.
- ❖ It is commonly used in foundations, load-bearing walls, retaining walls, industrial buildings, heavy-duty masonry, and other structures where high strength and durability are required.



Classification of Mortar

- ❖ Lightweight mortar is a type of mortar with a bulk density of less than 15 kN/m^3 (1500 kg/m^3), prepared using lightweight porous aggregates such as pumice, expanded perlite, vermiculite, cinder, or slag instead of ordinary sand.
- ❖ It has low self-weight, good thermal insulation, good sound insulation, and is easy to handle and apply, thereby reducing the dead load of the structure.
- ❖ It is commonly used for partition walls, non-load-bearing masonry, roof insulation, lightweight concrete blocks, and other lightweight construction works.



Classification of Mortar

- ❖ Cement mortar is prepared by mixing Portland cement, clean sand, and water in suitable proportions, commonly 1:3, 1:4, or 1:6.
- ❖ It is one of the most widely used mortars in modern construction because of its high compressive strength, durability, good water resistance, and quick setting.
- ❖ It provides a strong and durable bond between masonry units and is suitable for both structural and finishing works.
- ❖ It is commonly used in brick masonry, stone masonry, plastering, flooring, tile fixing, foundations, and other indoor and outdoor construction works.



Classification of Mortar

- ❖ Lime mortar is prepared by mixing lime, sand, and water, where the lime may be fat lime (non-hydraulic) or hydraulic lime.
- ❖ It provides excellent workability, flexibility, breathability, and good adhesion, with less shrinkage and cracking compared to cement mortar.
- ❖ It can also self-heal small cracks and is considered relatively environment-friendly.
- ❖ Lime mortar is mainly used in heritage structures, stone masonry, restoration work, traditional buildings, and structures where slight movement is expected.



Classification of Mortar

- ❖ Gypsum mortar is prepared using gypsum as the binding material and sets rapidly after mixing with water, providing a smooth and fine surface finish.
- ❖ It has good fire resistance, easy application, and excellent finishing properties, making it suitable for interior construction works.
- ❖ However, gypsum mortar is sensitive to moisture and is generally not recommended for damp areas or exterior applications.
- ❖ It is commonly used for interior plastering, decorative works, ceiling finishing, and dry indoor environments.



Classification of Mortar

- ❖ Mud mortar is prepared by mixing clay (mud) with water, with materials such as straw or cow dung sometimes added to improve strength and reduce cracking.
- ❖ It is one of the oldest and most economical types of mortar, as its materials are generally locally available and easy to prepare.
- ❖ However, it has low strength, poor durability, and low resistance to water, making it unsuitable for wet or heavily loaded structures.
- ❖ It is mainly used in rural houses, temporary buildings, traditional construction, and other low-cost structures.



Classification of Mortar

- ❖ Composite mortar is prepared by combining two or more binding materials, such as cement and lime or lime and surkhi, to obtain the advantages of different binders.
- ❖ It provides better workability, improved bond strength, reduced shrinkage, good durability, and greater economy compared with using a single binder alone.
- ❖ It is commonly used in residential buildings, commercial buildings, masonry work, plastering, and general construction works.
- ❖ It can be modified according to the specific requirements of strength, flexibility, and durability of the construction.



Classification of Mortar

- ❖ Masonry mortar is specially prepared to join bricks, stones, concrete blocks, and other masonry units into a strong and stable structure. It provides good bonding strength, adequate workability, high durability, and resistance to weathering, while also filling joints and distributing loads evenly throughout the masonry.
- ❖ It is commonly used in brick walls, stone walls, foundations, boundary walls, load-bearing structures, and general masonry construction.
- ❖ It also helps protect masonry joints from moisture penetration and environmental effects, thereby increasing the overall service life of the structure.



Classification of Mortar

- ❖ Plastering mortar is used to cover and finish the surfaces of walls and ceilings, providing a smooth, even, and attractive surface. It has good adhesion, easy workability, crack resistance, and good finishing properties, while also protecting masonry surfaces from moisture, weathering, dust, and mechanical damage.
- ❖ It is commonly used for wall plastering, ceiling plastering, decorative finishes, surface protection, and preparing surfaces for painting or other finishing treatments. It also improves the appearance and durability of the building by providing a uniform protective coating over the masonry surface.



Classification of Mortar

- ❖ **Fire-resistant mortar:** Used in furnaces, fireplaces, and chimneys to withstand high temperatures.
- ❖ **Waterproof mortar:** Used in water tanks, bathrooms, basements, and swimming pools to prevent water leakage.
- ❖ **Sound-absorbing mortar:** Used in auditoriums, recording studios, and theatres to reduce noise transmission.
- ❖ **X-ray shielding mortar:** Contains heavy aggregates such as barite and is used in hospitals and radiology rooms to block harmful radiation.
- ❖ **Chemical-resistant mortar:** Used in factories, laboratories, and chemical plants where resistance to acids and chemicals is required.
- ❖ **Polymer mortar:** Contains synthetic polymers to improve bonding strength, flexibility, and durability.



Ingredients of Mortar

Main Ingredients of Mortar

1. **Binding Material** – Cement, lime, or a combination of both, which provides strength.
2. **Fine Aggregate** – Clean, well-graded sand that adds bulk and reduces shrinkage.
3. **Water** – Initiates the chemical reaction and provides the required workability.
4. **Admixtures (Optional)** – Used to improve properties such as setting time, durability, or water resistance.

CONSTRUCTION MIX PROPORTIONS

(Table 12.1 Ref.)

MIX PROPORTIONS TABLE

Type of Work (w/ Icons)	Cement Part	Sand Part (w/ Icons)
 1. Masonry	1	4-5    
 2. Plastering	1	(a) 4 (Interior Icon)   (b) 5-6 (Exterior Icon)  
 3. Pointing	1	1-3   
 4. Reinforced Brick Work	1	3  
 5. Foundation	1	3-4   

Table 12.1



CEMENT MORTAR: Place within **30 MINUTES** after adding water.



MIXING OPTIONS: Choose **MANUAL MIXING** or **MECHANICAL MIXERS**.

LIME – CEMENT MORTAR

- Also known as guarded or gauged mortar.
- Prepared by mixing lime and cement.
- INCREASED BENEFITS:**
 -  Water Retention
 -  Workability
 -  Bonding Properties
 -  Frost Resistance

Mix Proportions (by Volume)

S.No.	Location	Ratio (by volume)				
		Cement	: Lime	:	Sand	
1.	Outside wall	1	:	1	:	6
		1	:	2	:	9
2.	Inside wall	1	:	2	:	9
		1	:	3	:	12



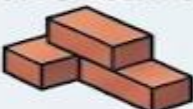
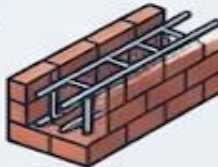
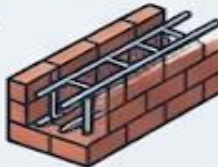
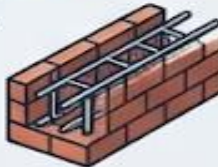
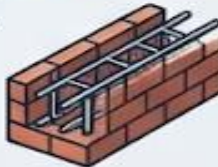
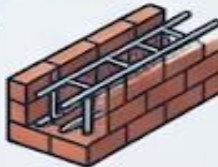
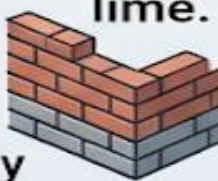
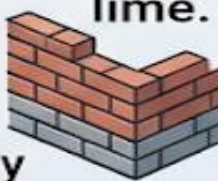
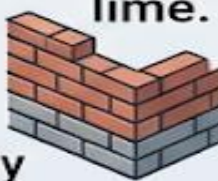
SELECTION OF MORTAR

GUIDE FOR CONSTRUCTION PROFESSIONALS

Nature of Work

Type of Mortar

Proportions

1.		Construction work in water-logged areas and exposed positions	Cement or lime mortar		1:3
2.		Damp-proof courses	Lime is eminently hydraulic		1:2
3.		General R.C.C. work such as (e.g., lintels, columns, stairs, etc.)	Cement mortar, the concrete Mix is 1:2:4		1:3
4.		Internal walls & surfaces of less importance	Lime cinder mortar. Sand is replaced by ashes or cinder.		1:3
5.		Mortar for laying fire-bricks	Fire-resisting mortar		1 part aluminous cement to 2 parts 2 fine fire-brick powder of fire-bricks.
6.		Partition walls & parapet walls	Cement mortar Lime mortar prop. 1:1.		1:3
7.		Plaster work	Lime moderately hydraulically hydraulic lime.		1:3 to 1:4
8.		Pointing work	Cement mortar		1:2
9.		Reinforced brickwork	Cement mortar		1:1 to 1:2
10.		Stone masonry with best varieties of stones	Cement mortar		1:3
11.		Stone masonry (ord. stones), stones, brickwork, foundations, etc.	Lime mortar		1:2
12.		Thin joints in brickwork	Lime mortar		1:6
			Lime should be eminently hydraulic lime.		1:2
			Lime: Eminently or moderately hydraulic lime.		1:3
			Lime being fat lime.		1:3



Characteristics of Mortar

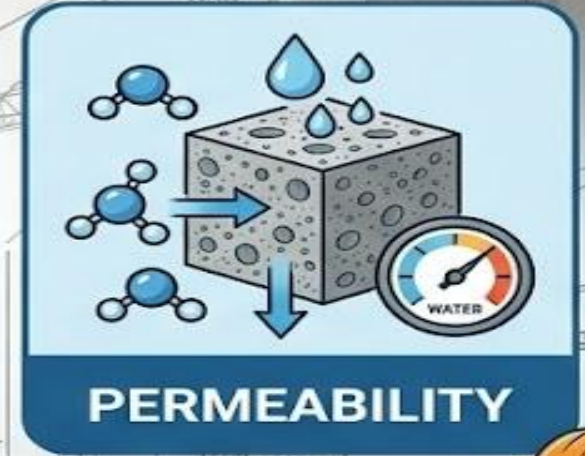
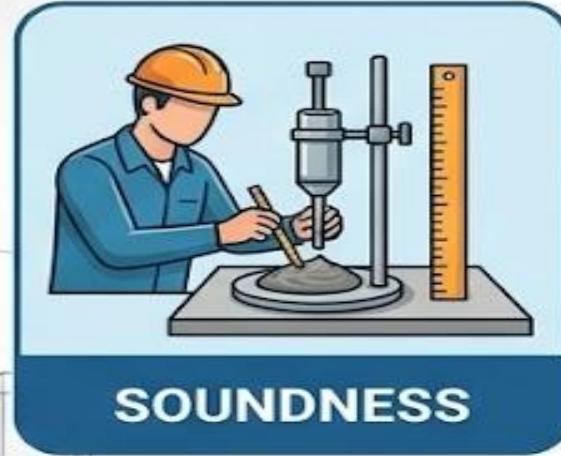
- **Workability** – It should be easy to mix, handle, spread, and place without excessive effort.
- **Good Adhesion** – It should bond properly with bricks, stones, blocks, or other masonry units.
- **Adequate Strength** – It should have sufficient compressive and tensile strength to safely transfer loads.
- **Durability** – It should resist weathering, moisture, temperature changes, and other environmental effects.
- **Water Retention** – It should retain enough water during and after application to allow proper hydration of cement and prevent premature drying.
- **Low Shrinkage** – It should have minimum shrinkage during drying to reduce cracking.
- **Good Cohesion** – The ingredients should remain uniformly mixed without segregation or bleeding.



Characteristics of Mortar

- **Impermeability** – Where required, it should resist the penetration of water and moisture.
- **Proper Setting Time** – It should remain workable long enough for placement but should set within a reasonable time.
- **Volume Stability** – It should maintain its volume after setting and should not undergo excessive expansion or contraction.
- **Economical** – It should provide the required performance at a reasonable cost using locally available materials.
- **Compatibility with Masonry Units** – Its strength and stiffness should be suitable for the bricks, blocks, or stones with which it is used.

TESTS ON MORTARS



RECOMMENDED PERMISSIBLE STRENGTH VALUES (N/mm²)

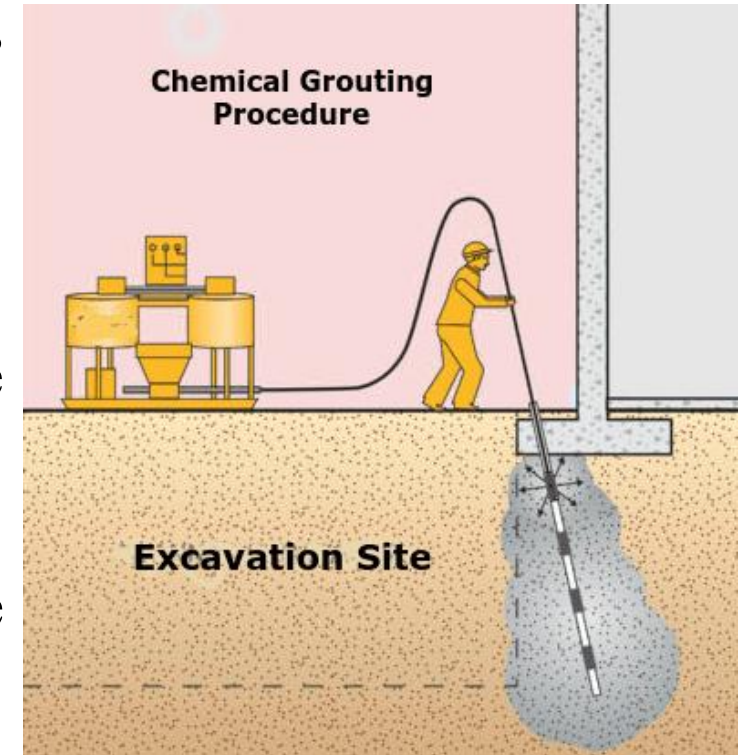
S.No.	Type of Mortar	Mix Proportion	Permissible strength (N/mm ²)
1.	Cement	1:3	0.75
2.	Cement	1:6	0.45
3.	Lime	1:3	0.45
4.	Cement-lime	1:1:6 or 1:1:9	0.50





Grout

- ❖ Cement mortar of fluid consistency used to fill the voids & joints in masonry & to repair the cracks is known as grout.
- ❖ Also used to increase SBC of soil by injection.
- ❖ Extensive use in dams, to fill the cracks formed after the concrete sets & hardens.
- ❖ Grout differs from mortar in its fluidity as it is to be poured & not to be spread into place with trowel.
- ❖ Composed of cement, sand, water , if needed grouting mixture.





Guniting

- ❖ Application of mortar or concrete under pneumatic pressure through a cement gun is known as guniting, concrete becomes extremely strong & a high bond is achieved.
- ❖ Composed of cement & sand conveyed through gun.
- ❖ Water is added at high velocity & the mix strikes the surface.



THANK

YOU